

Horizontal and vertical lines



1 State whether the graphs of the following lines are horizontal or vertical:

a $y = 2$

b $x = -4$

c $y = 3$

d $x = 17$

e $y = 0$

f $x = 0$

g $y - \frac{1}{2} = 0$

h $\frac{4}{7} = x$

2 The table shows some points on the line with equation $x = 0$.

Does the line $x = 0$ represent the y -axis or x -axis?

x	0	0	0	0
y	-8	-1	4	6

3 The table shows some points on the line with equation $y = 0$.

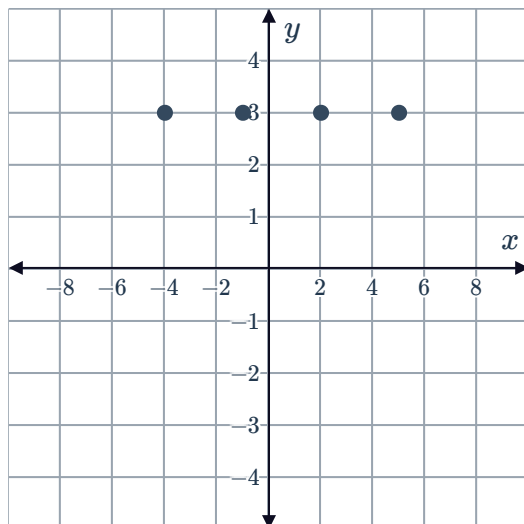
Does the line $y = 0$ represent the y -axis or x -axis?

x	-6	-4	1	5
y	0	0	0	0

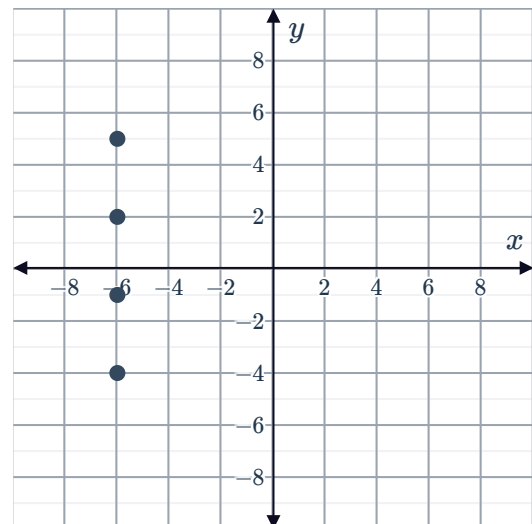
4 Consider the sets of points in the following coordinate planes:

- State whether the set of points lies on a increasing, decreasing, vertical or horizontal line.
- Find the equation of the line that passes through the set of points.

a



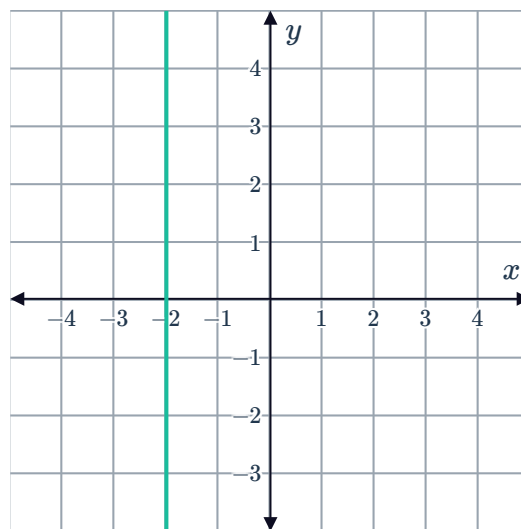
b



5 Consider the graphed line:

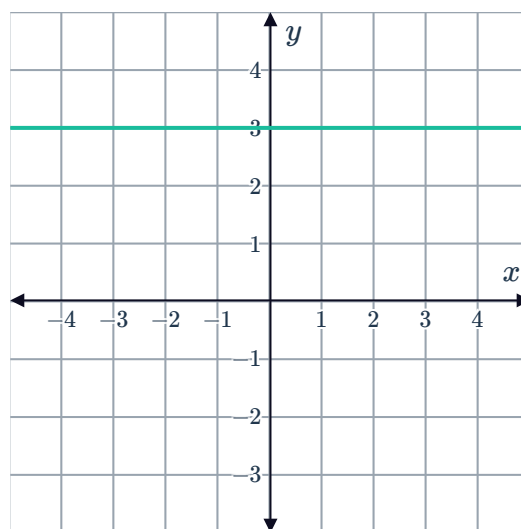


- a State the x -intercept.
- b Does this line have a y -intercept?
- c Write down the equation of the line.



6 Consider the graphed line:

- a State the y -intercept.
- b Does this line have an x -intercept?
- c Write down the equation of the line.



7 For each of the following tables of values:

- i Sketch the graph of the line that passes through the four points.
- ii Find the equation of the line.

a

x	2	2	2	2
y	-2	-1	0	1

b

x	-4	-2	0	2
y	5	5	5	5

8 Plot the following lines on a coordinate plane:

a $y = 8$

b $x = 7$

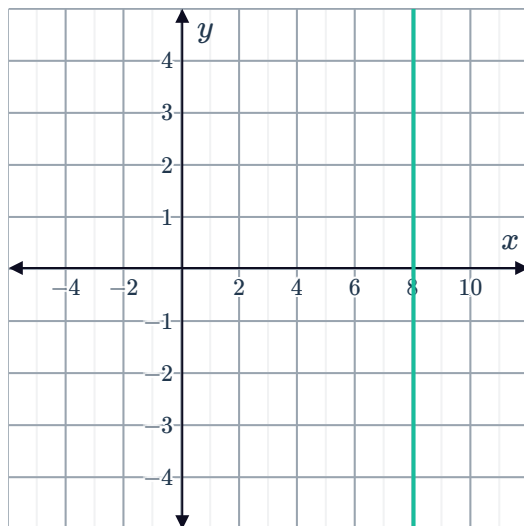
c $y = -2$

d $y = 0$

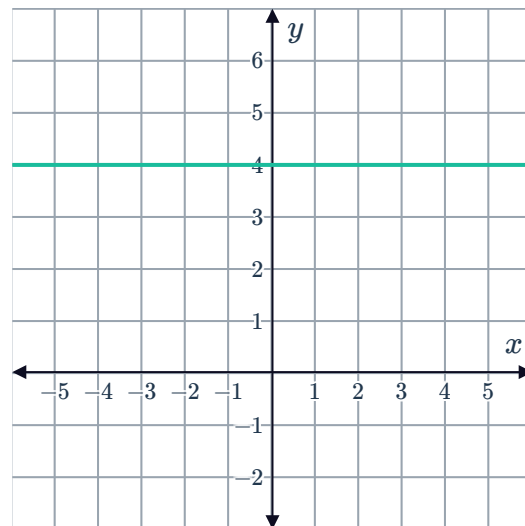
9 Find the equation of the following lines:



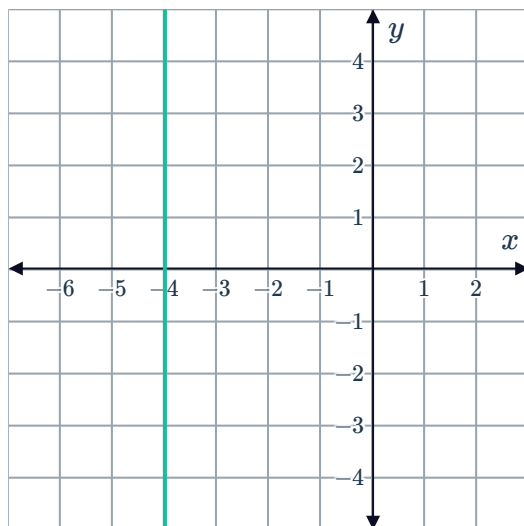
a



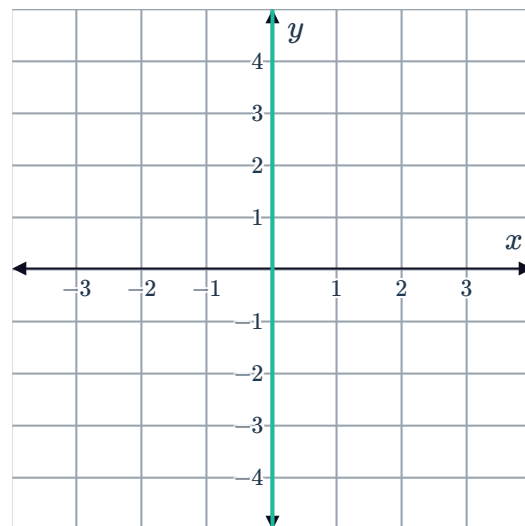
b



c



d



10 Find the equation of the line given the following conditions:

- a The line is parallel to the x -axis and passes through the point $(3, -5)$.
- b The line is parallel to the y -axis and passes through the point $(-8, 3)$.
- c The line is perpendicular to the x -axis and passes through the point $(3, -8)$.
- d The line is perpendicular to the y -axis and passes through the point $(3, -4)$.

- 11 The temperature (in degrees Celsius) inside a freezer over a five-hour period was recorded:



Time (h)	0	1	2	3	4	5
Temperature (T)	-9	-9	-9	-9	-9	-9

- a Sketch the line that passes through each of the pairs of values from the table.
- b Write an equation for the temperature T , at time h .